INTELLIGENT CUSTOMER CHURN PREDICTION

Using Data Analytics and Classification Models

A MINI PROJECT REPORT

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

K L (Deemed to be) University

Hyderabad, Telangana

Submitted by

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Under the guidance of

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Data Analytics and Visualization (24DSB3101)

Academic Year 2026–2027

DECLARATION

We hereby declare that the mini-project report entitled “Intelligent Customer Churn Prediction Using Data Analytics and Classification Models” is a record of work carried out by us in partial fulfillment of the requirements for the award of the Bachelor of Technology degree at K L (Deemed to be) University.

The project work, implementation details, dashboard captures and experimental results included in this report are based on the project repository and the execution evidence available to us. External datasets, research papers, software libraries and documentation are acknowledged in the references. We have not intentionally presented generated illustrations or fabricated experimental outputs as project evidence.

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CERTIFICATE

This is to certify that the mini-project report entitled “Intelligent Customer Churn Prediction Using Data Analytics and Classification Models” is a bona fide record of work carried out by Allipuram Akhil (2420030540) and Devineni Surya (2420030440), under the supervision of Dr. K. Swapnika, during the academic year 2026–2027, in partial fulfillment of the requirements for the award of the Bachelor of Technology degree at K L (Deemed to be) University.

Project Supervisor	Head of Department	External Examiner
Dr. K. Swapnika		

ACKNOWLEDGEMENT

We express our sincere gratitude to our faculty guide, Dr. K. Swapnika, for her guidance, suggestions and support throughout the development of this project. Her feedback helped us structure the data analysis, model evaluation and project documentation.

We thank the Department of Computer Science and Engineering, K L (Deemed to be) University, for providing the academic environment and resources required to complete the Data Analytics and Visualization project.

We also acknowledge the maintainers of the public Telco Customer Churn dataset, the developers of Python and the machine-learning libraries used in this implementation, and the authors of the research works and technical documentation cited in this report.

Finally, we thank our families, classmates and everyone who provided assistance during the project.

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1. ABSTRACT

Customer churn is a significant business concern for subscription-based telecommunications services because customers who discontinue a service reduce recurring revenue and may increase replacement-acquisition costs. This project develops an end-to-end customer churn prediction workflow using the publicly circulated IBM Telco Customer Churn sample containing 7,043 customer records and 21 columns. The target variable is binary: churn or no churn. The data preparation stage converts TotalCharges to numeric form, handles 11 blank values associated with zero-tenure records, excludes the customer identifier from model inputs, and encodes the target. Exploratory analysis is used to examine churn patterns across contract type, internet service, payment method, tenure and charges. A stratified 80:20 train-test split with random state 42 is used, and preprocessing is placed inside a scikit-learn pipeline to reduce leakage risk. The dashboard repository includes Logistic Regression, Decision Tree, Random Forest, XGBoost, RBF Support Vector Machine, an MLP neural network and a Random Forest–XGBoost soft-voting hybrid. In the recorded dashboard evaluation, the hybrid obtains ROC-AUC 0.8439, PR-AUC 0.6510, F1 0.6327 and recall 0.7647. It leads on ROC-AUC and PR-AUC among the displayed models, while the MLP has the highest accuracy (0.7977) and precision (0.6353). The Streamlit application provides overview, EDA, model comparison, evaluation, single-customer prediction, batch CSV prediction and explainability pages. These outputs support exploratory risk assessment; they do not establish causal relationships or guarantee future churn.

Keywords: Customer churn, telecom analytics, binary classification, Random Forest, XGBoost, ensemble learning, Streamlit, SHAP.

Team details and responsibilities

Member	Roll number	Project contribution
Allipuram Akhil	2420030540	Project integration, data analysis, model comparison, dashboard and documentation
Devineni Surya	2420030440	Data preparation, evaluation support, testing and documentation

Dataset: WA_Fn-UseC_-Telco-Customer-Churn.csv (7,043 rows × 21 columns). Repository: customer-churn-dav-stage8 project archive supplied for this work.

2. INTRODUCTION

Telecommunication providers operate in a competitive subscription market. Customers can change providers or discontinue services, and churn can affect recurring revenue, customer lifetime value and retention planning. A churn prediction system estimates whether a customer record resembles the historical profile of customers who left. This estimate can help an organization prioritize customer-retention review, provided that the score is interpreted with business context.

The project treats churn prediction as a supervised binary-classification task. The input is a structured customer record containing demographic attributes, subscribed services, account information, contract details and billing information. The output is a churn probability and a class label. The target field is Churn, encoded as 1 for churn and 0 for no churn.

The dataset contains 7,043 customer records: 1,869 churners (26.54%) and 5,174 non-churners (73.46%). This imbalance means accuracy alone is insufficient. Recall, precision, F1, specificity, ROC-AUC, PR-AUC and other metrics are therefore used to describe different aspects of classification performance.

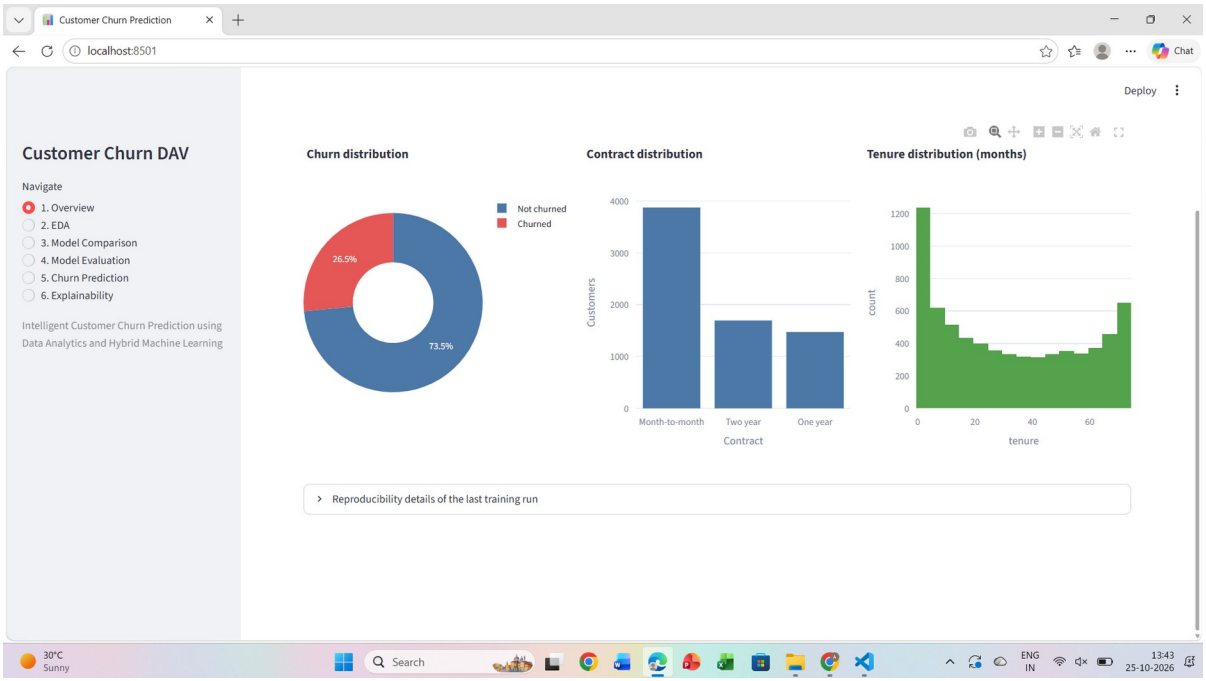


Figure 1. Original Streamlit overview page showing churn distribution, contract distribution and tenure distribution.

The implemented workflow includes data loading and cleaning, exploratory data analysis, feature engineering, training and comparison of baseline classifiers and a proposed hybrid, held-out evaluation, explainability and a Streamlit interface. The report follows the KLH mini-project documentation structure: literature survey, problem definition, requirements, implementation, results, conclusion and references.

3. LITERATURE SURVEY

The project literature register identifies research and technical sources relevant to churn prediction, explainability, data leakage and probability calibration. The summaries below are limited to the recorded source descriptions; metrics from unrelated datasets are not treated as directly comparable with this project.

3.1 Selected research and technical sources

Imani et al. (2025), systematic review of customer churn prediction. The review surveys recent machine-learning and deep-learning approaches to customer churn prediction. Its relevance is the broad context it provides for ensemble methods, class imbalance, interpretability, concept drift and business-aligned evaluation. As a review, it does not demonstrate that the model used in this project is superior.

“Explaining customer churn prediction in telecom industry using tabular machine learning models” (2024). The recorded literature register identifies this work as relevant to telecom churn prediction using tabular machine-learning models and SHAP-based explanation. It supports the use of explainability to inspect model behaviour. Full author and publication metadata should be checked against the publisher record before formal submission.

“A data-driven approach with explainable artificial intelligence for customer churn prediction in the telecommunications industry” (2025). This recent work is listed in the project register as an example of explainable AI applied to telecom churn prediction. It provides current domain context, but results from a different dataset and experimental protocol should not be compared numerically with the present project.

Lundberg and Lee (2017), SHAP. The NeurIPS paper introduces a unified approach to interpreting model predictions using Shapley-value-based feature attribution. The project uses SHAP summary and mean absolute SHAP plots to inspect which inputs influence the fitted hybrid model. These explanations describe model associations, not causal effects.

IBM Telco Customer Churn sample documentation. IBM documentation provides context for the Telco churn sample. The project uses the commonly circulated single CSV with 7,043 rows and 21 columns; it should not be assumed to be identical to every multi-table version of the IBM sample.

Scikit-learn documentation on common pitfalls and data leakage. The documentation explains why preprocessing and feature transformations should be fitted only on training data. The project places imputation, scaling and categorical encoding inside a pipeline so that the transformations are learned from the training partition.

Scikit-learn probability calibration documentation. The documentation describes calibration curves and probability quality. The project includes a reliability analysis as an exploratory diagnostic and notes that class weighting can affect the interpretation of raw predicted probabilities.

3.2 Research motivation and project-level gap

The reviewed sources show that churn prediction and explainable machine learning are established research areas. The project therefore does not claim to be the first to combine ensemble learning and explainability. Its practical motivation is to provide a reproducible comparison of several classifiers and a Random Forest–XGBoost soft-voting model on one documented benchmark, using multiple threshold-dependent and threshold-independent metrics and an interactive dashboard. This is a scoped project question, not a claim of research novelty.

4. PROBLEM STATEMENT AND OBJECTIVES

4.1 Problem statement

Develop a data-driven classification system that analyses historical telecom customer attributes and estimates the likelihood that a customer will discontinue the service, presenting the prediction and supporting evaluation evidence through an interactive dashboard.

4.2 Problem description

Manual inspection of thousands of customer records is time-consuming and makes it difficult to consistently identify combinations of contract, service and billing characteristics associated with churn. The dataset is imbalanced, so a model that only maximizes accuracy may fail to identify a useful share of churners. The system therefore compares models using multiple metrics and presents its output as a risk estimate rather than a certainty.

4.3 Objectives

- Load and inspect the 7,043-row Telco Customer Churn dataset and document its target distribution.
- Clean the TotalCharges field, handle blank values and exclude the customer identifier from model inputs.
- Perform exploratory analysis of churn by contract, payment method, internet service, tenure, charges and customer attributes.
- Create additional customer-level features and apply preprocessing within a training-fitted pipeline.
- Train and compare Logistic Regression, Decision Tree, Random Forest, XGBoost, RBF SVM, MLP and the RF–XGBoost hybrid as represented in the dashboard outputs.
- Evaluate models using accuracy, precision, recall, specificity, F1, ROC-AUC, PR-AUC, MCC, Cohen’s Kappa, log loss and confusion matrices.
- Provide model explainability views and an interactive Streamlit interface for single-customer and batch CSV prediction.
- Document limitations, including the single public dataset, class imbalance, probability calibration and lack of external validation.

5. HARDWARE AND SOFTWARE REQUIREMENTS

5.1 Hardware requirements

- A laptop or desktop computer with a modern multi-core CPU.
- At least 8 GB RAM recommended for data processing, model training and dashboard execution.
- Sufficient disk space for the CSV dataset, Python packages, saved models, figures and prediction outputs.
- Internet access for obtaining packages and consulting dataset or library documentation. A dedicated GPU is not required for the recorded classical tabular-model workflow.

5.2 Software requirements

Component	Used / specified in project
Operating system	Windows environment used for dashboard screenshots; Python is cross-platform
Programming language	Python 3.12.3 in the recorded run metadata
Data handling	pandas, NumPy
Machine learning	scikit-learn, XGBoost
Explainability	SHAP
Visualization	Matplotlib / Plotly and project visualization modules
Dashboard	Streamlit
Data format	CSV
Version control / project structure	Git-compatible repository structure

5.3 Main repository modules

Area	Repository path / artifact	Purpose
Application entry	app.py, dashboard/app.py	Launches the Streamlit application
Data loading and cleaning	src/data/	Loads raw data and prepares clean records
Feature engineering	src/features/	Creates derived customer-level features
Models	src/models/	Defines baseline and hybrid classifiers
Evaluation	src/evaluation/	Computes metrics, comparisons and error analysis
Explainability	src/explainability/	Runs SHAP-based analysis
Saved results	outputs/metrics/, outputs/figures/	Stores tables, plots and predictions

6. IMPLEMENTATION

6.1 Dataset and task

The project uses the WA_Fn-UseC_-Telco-Customer-Churn.csv file in the supplied repository. It contains 7,043 rows and 21 columns, including customerID, the target Churn, and 19 customer attributes. The attributes cover demographics, service subscriptions, tenure, contract type, billing method and charges.

Dataset property	Recorded value
Records	7,043
Columns	21
Churn = Yes	1,869 (26.54%)
Churn = No	5,174 (73.46%)
Target	Churn (binary)
Identifier handling	customerID excluded from model features

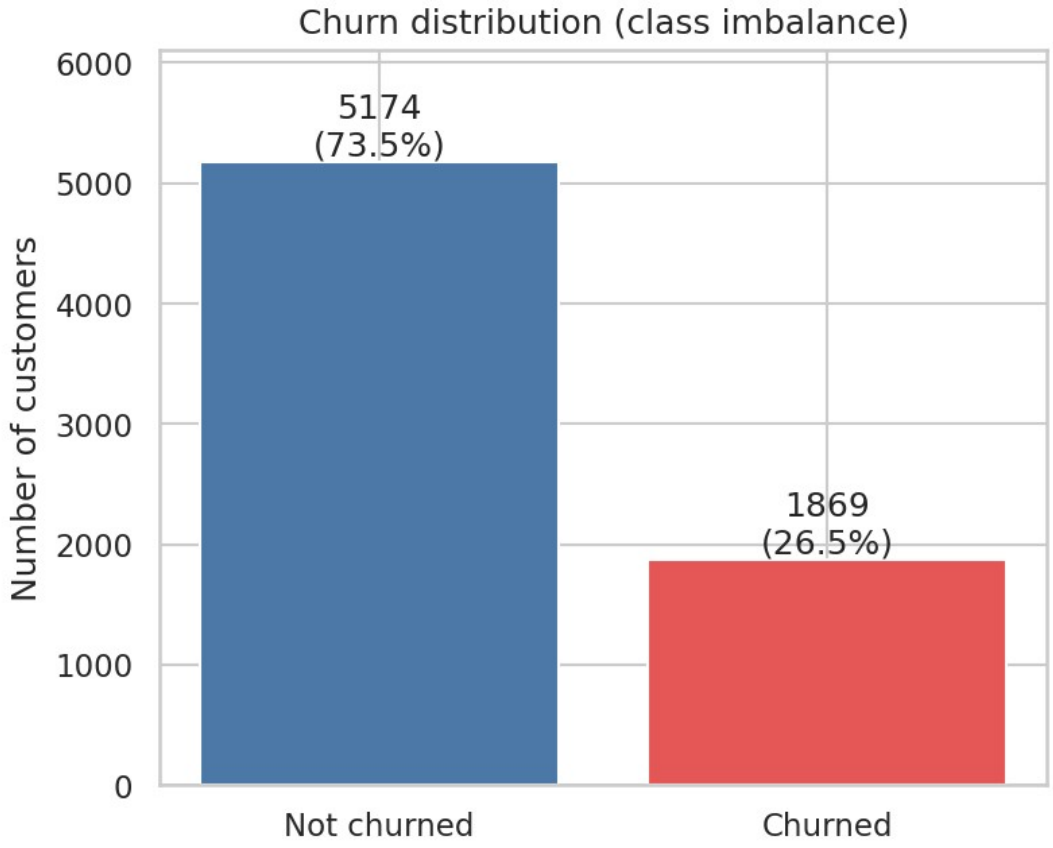


Figure 2. Original project-generated churn distribution chart from outputs/figures.

6.2 Data cleaning and preprocessing

- TotalCharges is initially represented as text and is converted to numeric.
- The 11 blank TotalCharges values occur in records with tenure equal to zero and are converted to 0.0.
- The project reports zero exact duplicate rows and zero duplicate customer IDs in its review content.
- customerID is excluded from model features; Churn is mapped from Yes/No to 1/0.
- The project reports no rows removed under the 1.5 × IQR outlier check for tenure, MonthlyCharges and TotalCharges.
- Numeric columns use median imputation and standardization; categorical columns use mode imputation and one-hot encoding.
- The preprocessing transformations are included inside the scikit-learn pipeline and fitted using training data only.

6.3 Train-test split and validation

A stratified 80:20 split is used with random_state = 42. The recorded split contains 5,634 training records and 1,409 test records. Stratification preserves the approximate class proportions. Five-fold stratified cross-validation is performed on the training partition. The test partition is used for the reported holdout evaluation.

6.4 Exploratory data analysis

The dashboard and project-generated figures examine churn distribution and relationships with contract type, tenure, monthly charges, services, payment method and internet service. These are descriptive associations and should not be interpreted as proof that a particular service or contract causes churn.

6. IMPLEMENTATION (CONTINUED)

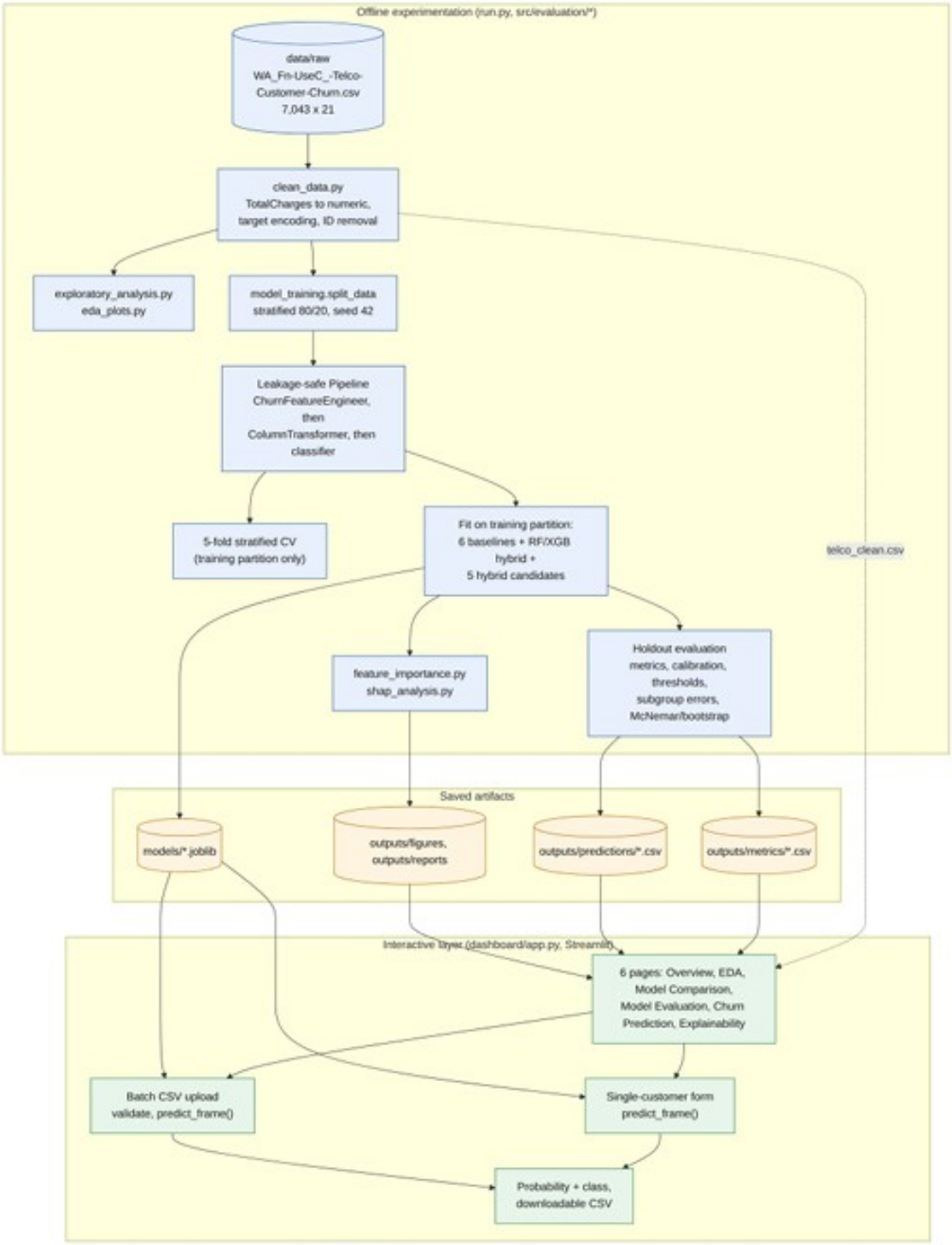


Figure 3. Original repository-traced system architecture supplied by the project team. Blue: offline experimentation; orange: saved artifacts; green: interactive Streamlit layer.

The architecture diagram shows the offline pipeline beginning with the raw Telco CSV. Cleaning and train-test splitting feed a leakage-safe preprocessing pipeline. Model fitting and evaluation produce saved model files, figures, reports, prediction CSV files and metric CSV files. The Streamlit dashboard reads these saved artifacts and exposes six pages: Overview, EDA, Model Comparison, Model Evaluation, Churn Prediction and Explainability. Single-customer and batch CSV flows produce churn probability and class outputs.

6.5 Feature engineering

The project documents 12 derived features: tenure_group, service_count, protection_service_count, streaming_service_count, avg_monthly_charge, charge_delta, charges_per_service, is_month_to_month, auto_payment, is_electronic_check, has_family and mtm_fiber. The features summarize tenure, service usage, billing behaviour and selected interactions. The project documentation describes these as row-wise transformations, with no target-derived value used to create the feature.

6.6 Baseline models and proposed hybrid

Model	Role / configuration summary
Logistic Regression	Interpretable linear benchmark; class_weight balanced
Decision Tree	Non-linear rule-based benchmark; max depth 5, min leaf 20
Random Forest	Bagging ensemble; 300 trees, max depth 10, min leaf 5
XGBoost	Gradient-boosted trees; 300 estimators, learning rate 0.05, depth 4
RBF SVM	Kernel-based classifier included in the dashboard comparison
MLP neural network	Feed-forward neural-network classifier included in dashboard comparison
RF + XGBoost hybrid	Weighted soft voting; default probability average 0.5 × RF + 0.5 × XGBoost

For the hybrid, the probability is computed as P_hybrid = 0.5 × P_RF + 0.5 × P_XGBoost. A customer is classified as churn when the hybrid probability is at least 0.5. The weights are configurable in the implementation. The project does not claim that the hybrid is universally superior; performance is compared metric by metric.

6. IMPLEMENTATION (CONTINUED)

6.7 Dashboard implementation

The Streamlit dashboard presents six navigable pages. The Overview page summarizes the dataset. EDA presents interactive charts for selected customer variables. Model Comparison displays a held-out test-set metric table and a best-model-per-metric view. Model Evaluation displays a selected model's confusion matrix, classification metrics, ROC curve and precision-recall curve. Churn Prediction provides a form for a single customer and a CSV upload route for batch scoring. Explainability presents permutation importance and SHAP outputs.

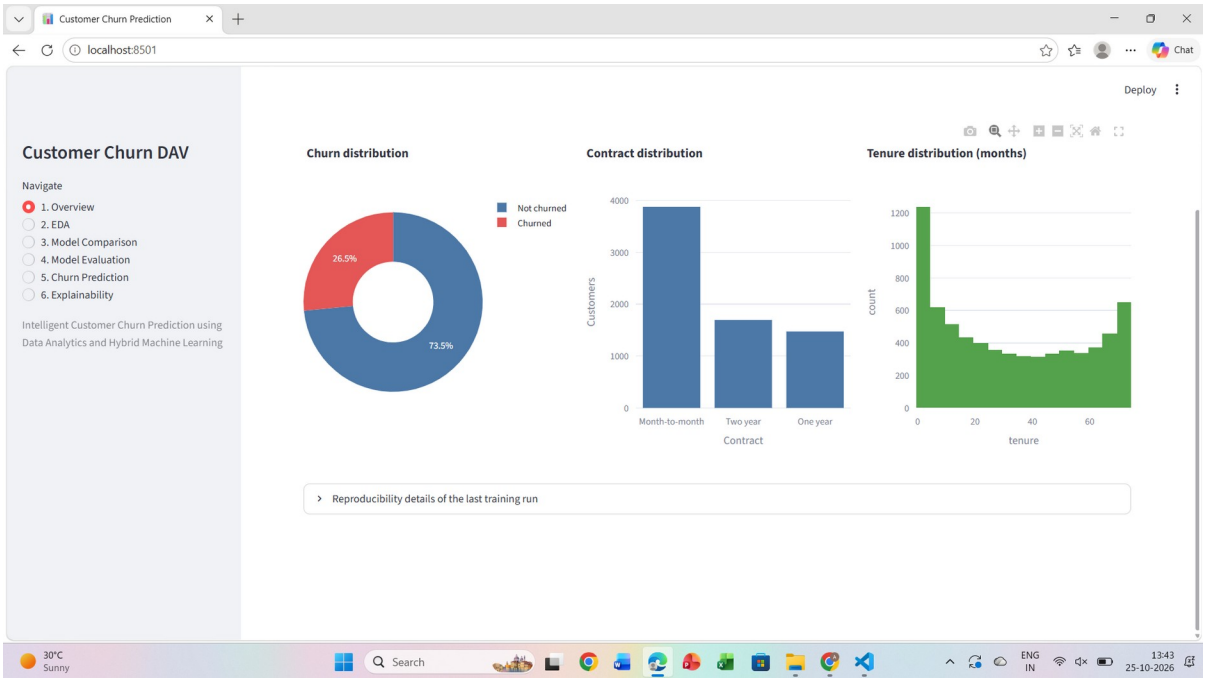
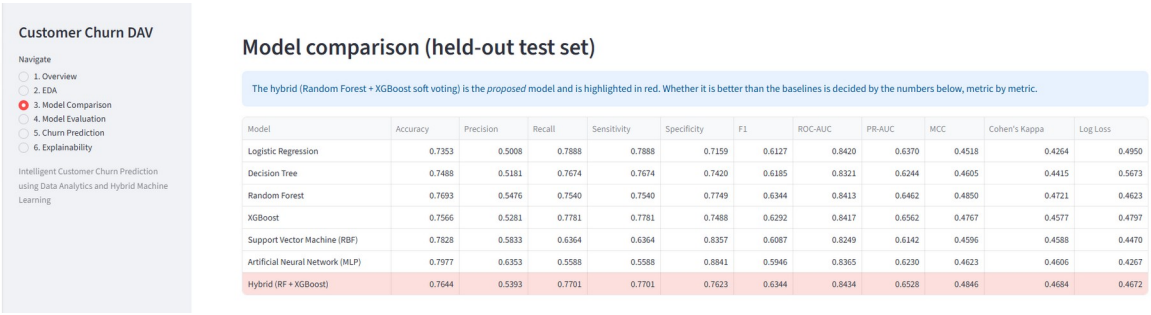


Figure 4. Original dashboard overview screenshot.



Model	Accuracy	Precision	Recall	Sensitivity	Specificity	F1	ROC-AUC	PR-AUC	MCC	Cohen's Kappa	Log Loss
Logistic Regression	0.7353	0.5008	0.7888	0.7888	0.7159	0.6127	0.8420	0.6370	0.4518	0.4264	0.4950
Decision Tree	0.7488	0.5181	0.7674	0.7674	0.7420	0.6185	0.8321	0.6244	0.4605	0.4415	0.5673
Random Forest	0.7693	0.5476	0.7540	0.7540	0.7749	0.6344	0.8413	0.6462	0.4850	0.4721	0.4623
XGBoost	0.7566	0.5281	0.7781	0.7781	0.7488	0.6292	0.8417	0.6562	0.4767	0.4577	0.4797
Support Vector Machine (RBF)	0.7828	0.5833	0.6364	0.6364	0.8357	0.6087	0.8249	0.6142	0.4596	0.4588	0.4470
Artificial Neural Network (MLP)	0.7977	0.6353	0.5588	0.5588	0.8841	0.5946	0.8365	0.6230	0.4623	0.4606	0.4267
Hybrid (RF + XGBoost)	0.7644	0.5393	0.7701	0.7701	0.7623	0.6344	0.8434	0.6528	0.4846	0.4684	0.4672

Figure 5. Original dashboard model-comparison table screenshot. Values shown are the captured dashboard run.

6.8 Single-customer prediction

The single-customer form collects the customer attributes used by the model, including gender, senior-citizen status, partner and dependent indicators, service subscriptions, tenure, contract, payment method and charges. On submission, the application displays a churn probability, a predicted class and a gauge-style probability visualization. The displayed probability is a model estimate, not a guarantee that the customer will leave.

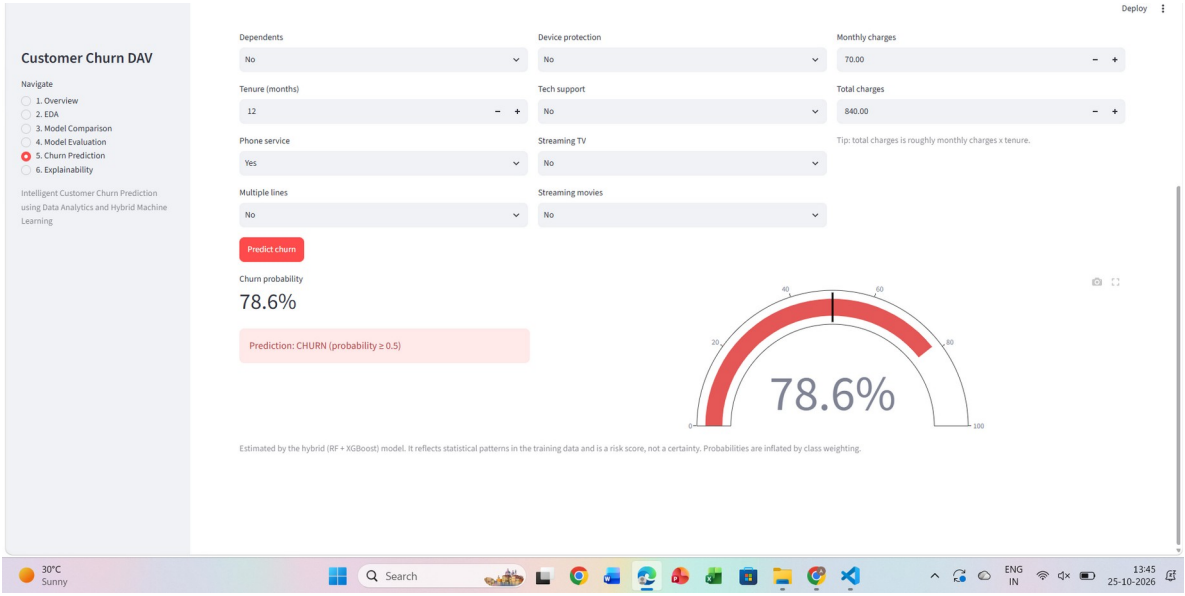


Figure 6. Original single-customer prediction screenshot showing a 78.6% churn probability for the displayed input record.

6.9 Batch CSV prediction

The batch workflow accepts a CSV with the required customer feature columns, validates the expected fields and scores each row. The displayed output includes customer_id, churn_probability and churn_prediction, with a download option for the scored CSV.

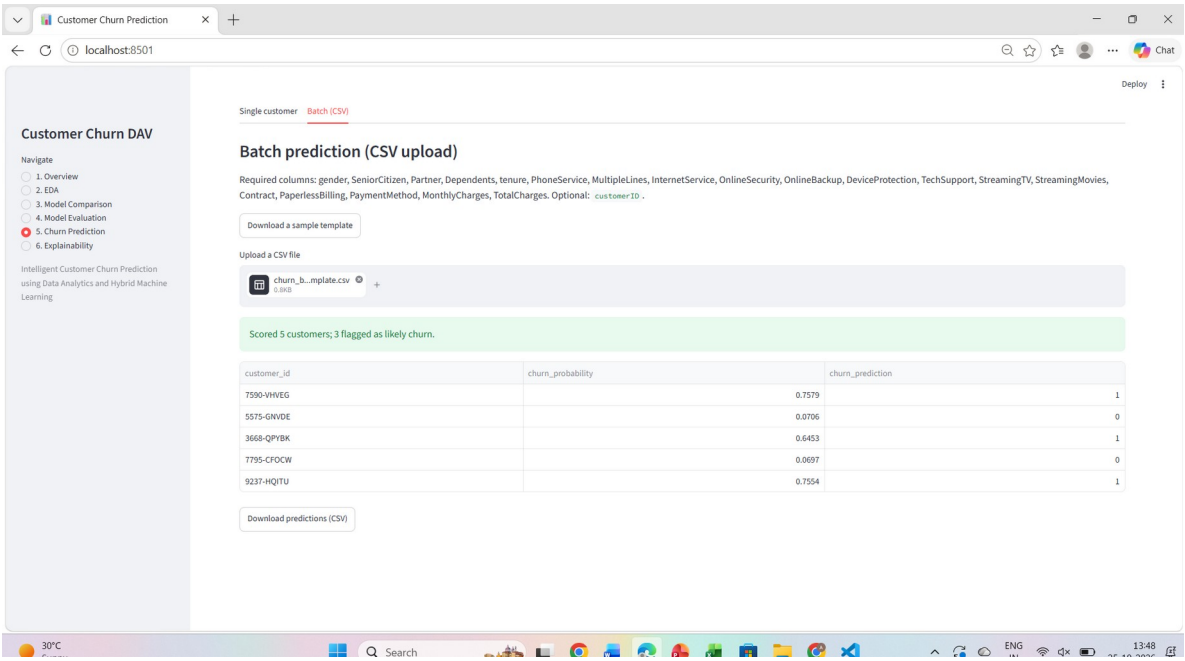


Figure 7. Original batch CSV prediction screenshot showing the uploaded file and scored customer rows.

7. RESULTS AND DISCUSSION

7.1 Exploratory findings

Segment	Observed churn rate
Month-to-month contract	42.7%
One-year contract	11.3%
Two-year contract	2.8%
Fiber-optic internet	41.9%
DSL internet	19.0%
No internet service	7.4%
Electronic-check payment	45.3%
Tenure 0–12 months	47.4%
Tenure 61+ months	6.6%
Senior citizens	41.7%
Non-senior customers	23.6%

The observed churn rates differ across contract and service groups. The largest displayed rates include month-to-month contracts, fiber-optic internet and electronic-check payment. The project also reports median tenure of 10 months for churners and 38 months for non-churners, and median monthly charges of 79.65 and 64.43 respectively. These values describe the dataset and do not establish causal effects.

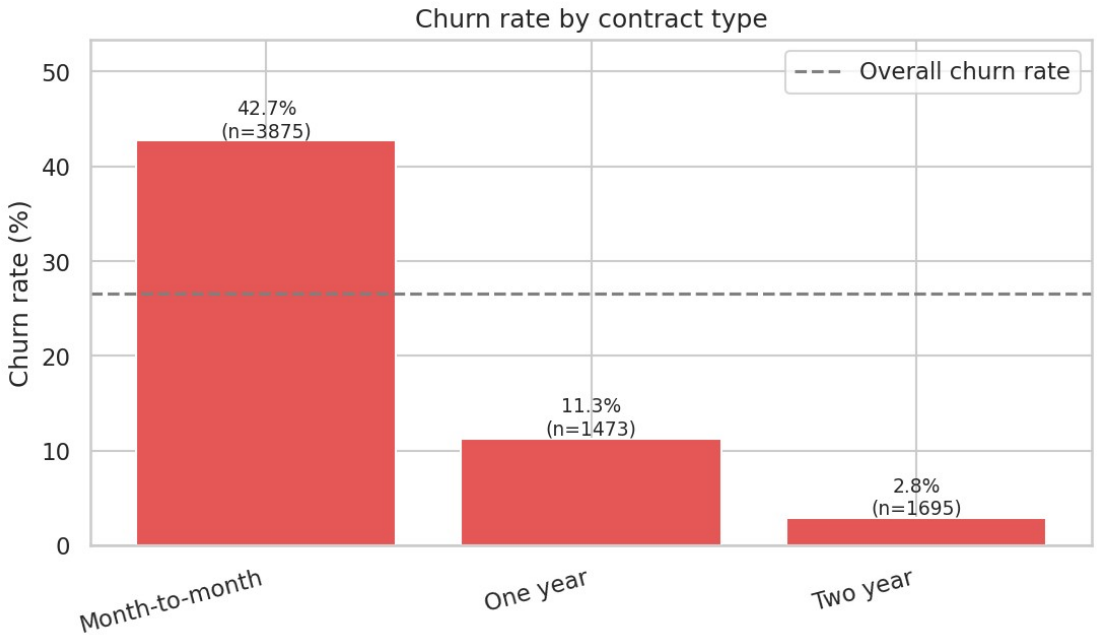


Figure 8. Project-generated churn rate by contract type.

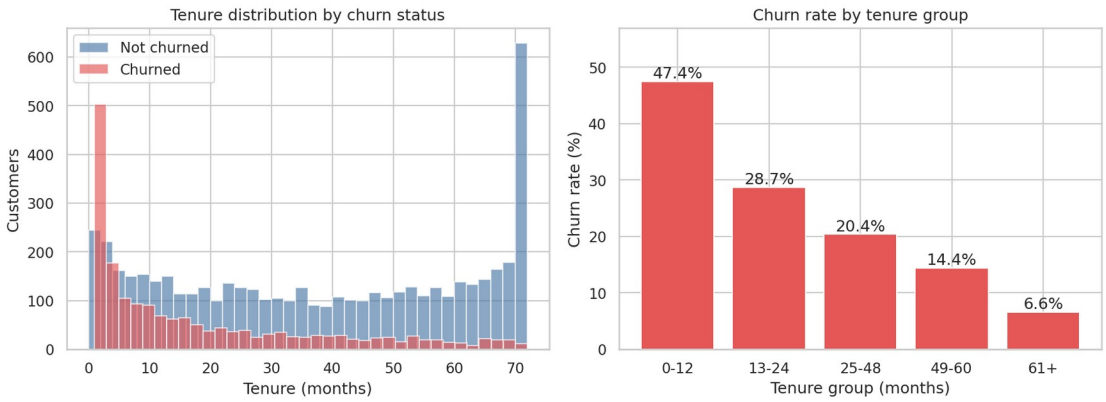


Figure 9. Project-generated tenure distribution and churn rate by tenure group.

7. RESULTS AND DISCUSSION (CONTINUED)

7.2 Model comparison

The following table reproduces the recorded dashboard comparison from the supplied project execution evidence. Metrics are rounded to four decimal places. The positive class is churn.

Model	Accuracy	Precision	Recall	Specificity	F1	ROC-AUC	PR-AUC
Logistic Regression	0.7353	0.5008	0.7888	0.7159	0.6127	0.8420	0.6370
Decision Tree	0.7488	0.5181	0.7674	0.7420	0.6185	0.8321	0.6244
Random Forest	0.7693	0.5476	0.7540	0.7749	0.6344	0.8413	0.6462
XGBoost	0.7566	0.5281	0.7781	0.7488	0.6292	0.8417	0.6562
RBF SVM	0.7828	0.5833	0.6364	0.8357	0.6087	0.8249	0.6142
MLP neural network	0.7977	0.6353	0.5588	0.8841	0.5946	0.8365	0.6230
Hybrid RF + XGBoost	0.7644	0.5393	0.7701	0.7623	0.6344	0.8434	0.6528

In the captured comparison, the hybrid has the highest ROC-AUC (0.8434) among the displayed models, while XGBoost has the highest PR-AUC (0.6562). The MLP has the highest accuracy (0.7977), precision (0.6353) and specificity (0.8841), but its recall is lower (0.5588). The hybrid F1 score is 0.6344, matching the Random Forest value at the displayed precision. No single model leads on every metric.

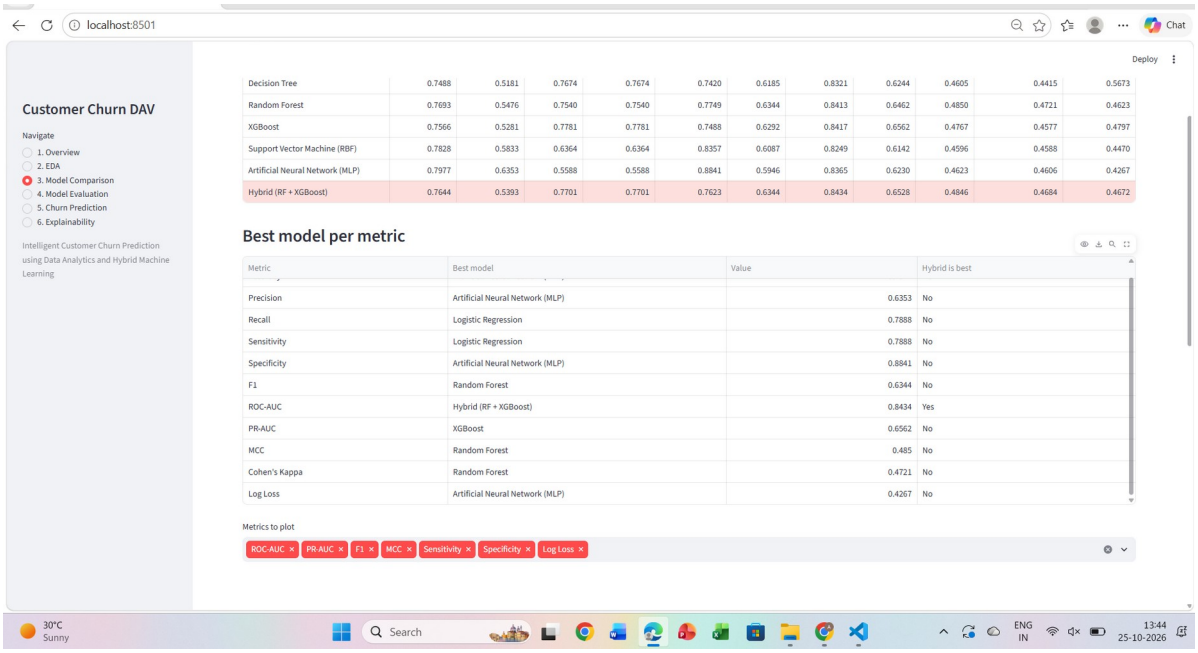


Figure 10. Original dashboard best-model-per-metric table and selected metrics view.

7. RESULTS AND DISCUSSION (CONTINUED)

7.3 Confusion matrix and curves

The captured hybrid evaluation reports 789 true negatives, 246 false positives, 86 false negatives and 288 true positives. Therefore, 288 of the 374 actual churners in the captured test partition are identified, corresponding to recall of approximately 0.7701. The 246 false positives represent non-churners classified as churn. The result should be interpreted with the class imbalance and decision threshold in mind.

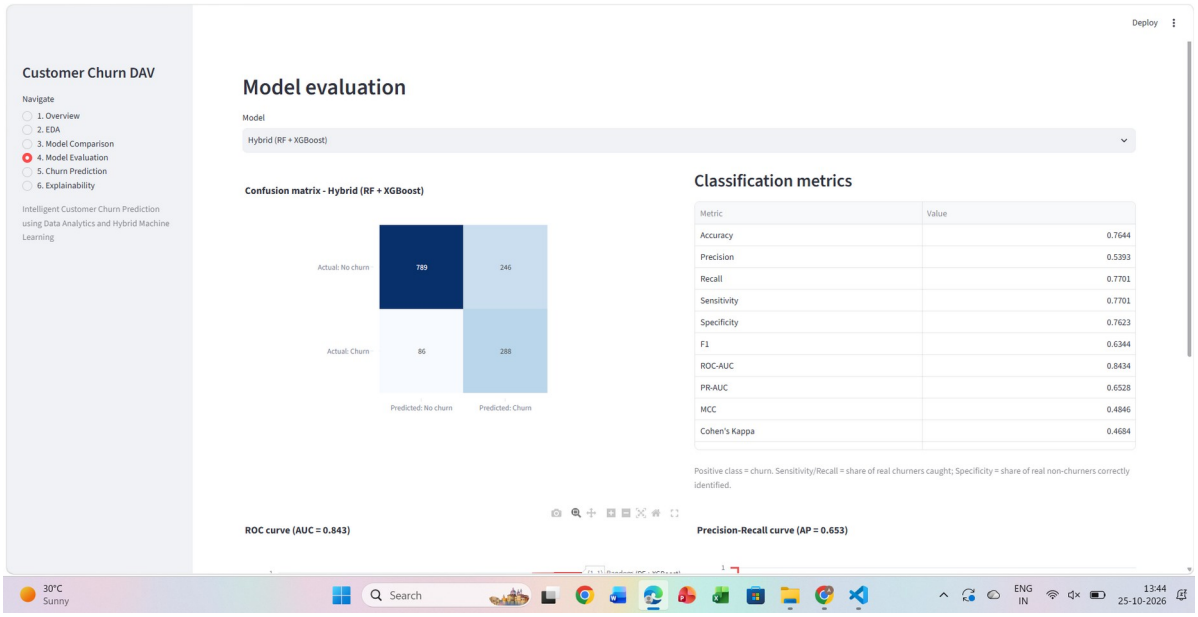


Figure 11. Original dashboard hybrid confusion matrix and classification metrics.

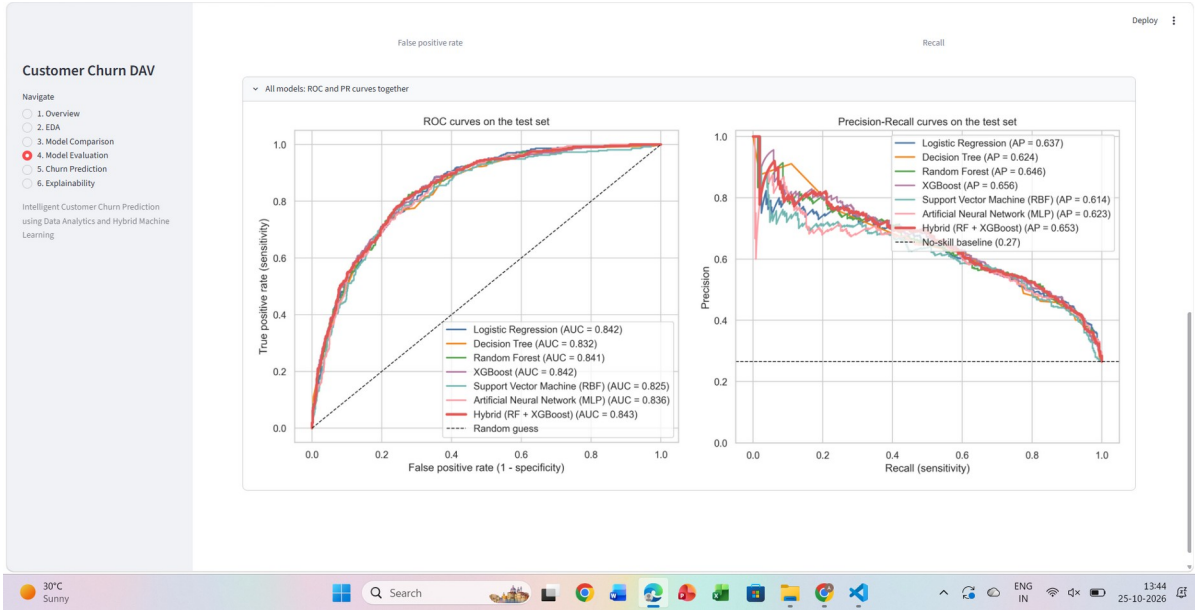


Figure 12. Original dashboard ROC and precision-recall curves for the hybrid and comparison models.

7.4 Explainability

The explainability page includes permutation importance and SHAP-based summaries. In the captured dashboard, contract type, tenure and internet-service-related features appear among the higher-ranked variables. Other displayed features include total charges, online security, payment method, tech support and monthly charges. These plots indicate how the fitted model uses input features; they do not demonstrate that the variables cause customer churn.



Figure 13. Original dashboard permutation-importance chart for the hybrid model.

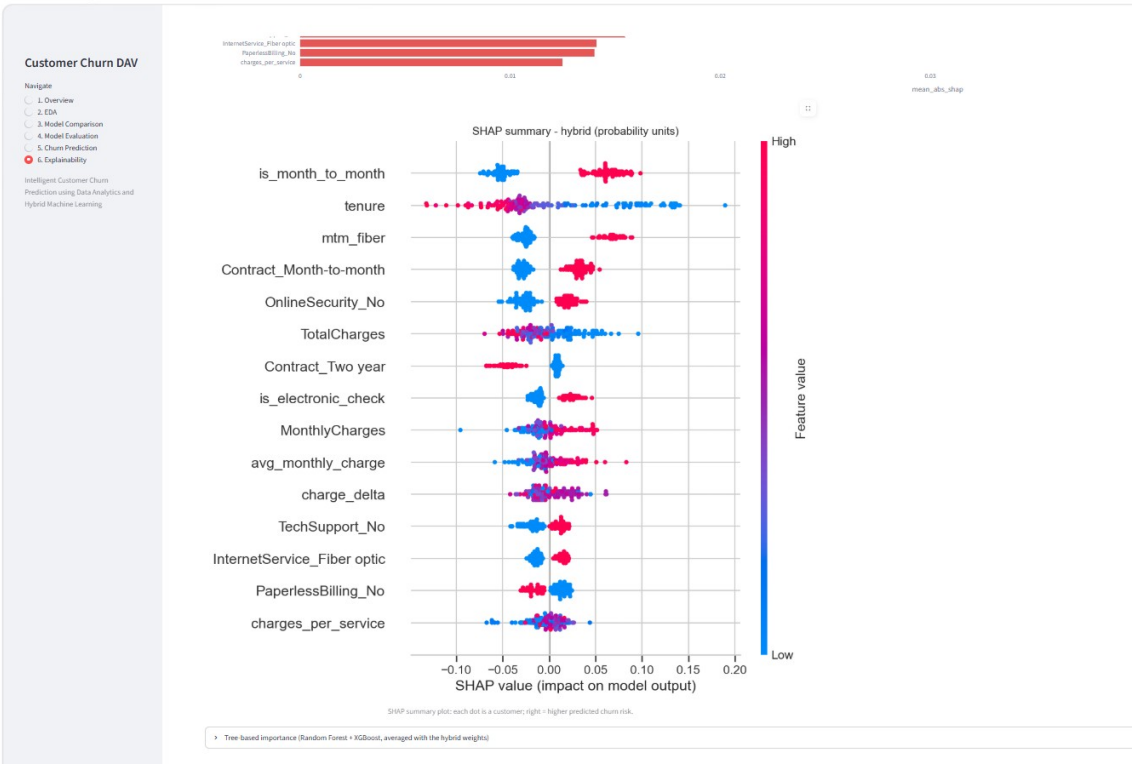


Figure 14. Original dashboard SHAP summary plot for the hybrid model.

7. RESULTS AND DISCUSSION (CONTINUED)

7.5 Additional evaluation and interpretation

The project archive also contains cross-validation, calibration, threshold and error-analysis artifacts. These outputs are intended to examine variability and the behaviour of predicted probabilities beyond a single accuracy figure. The calibration and threshold figures should be treated as diagnostic evidence from the recorded run, not as proof that the probabilities are fully calibrated or that the selected 0.5 threshold is optimal for a telecom retention programme.

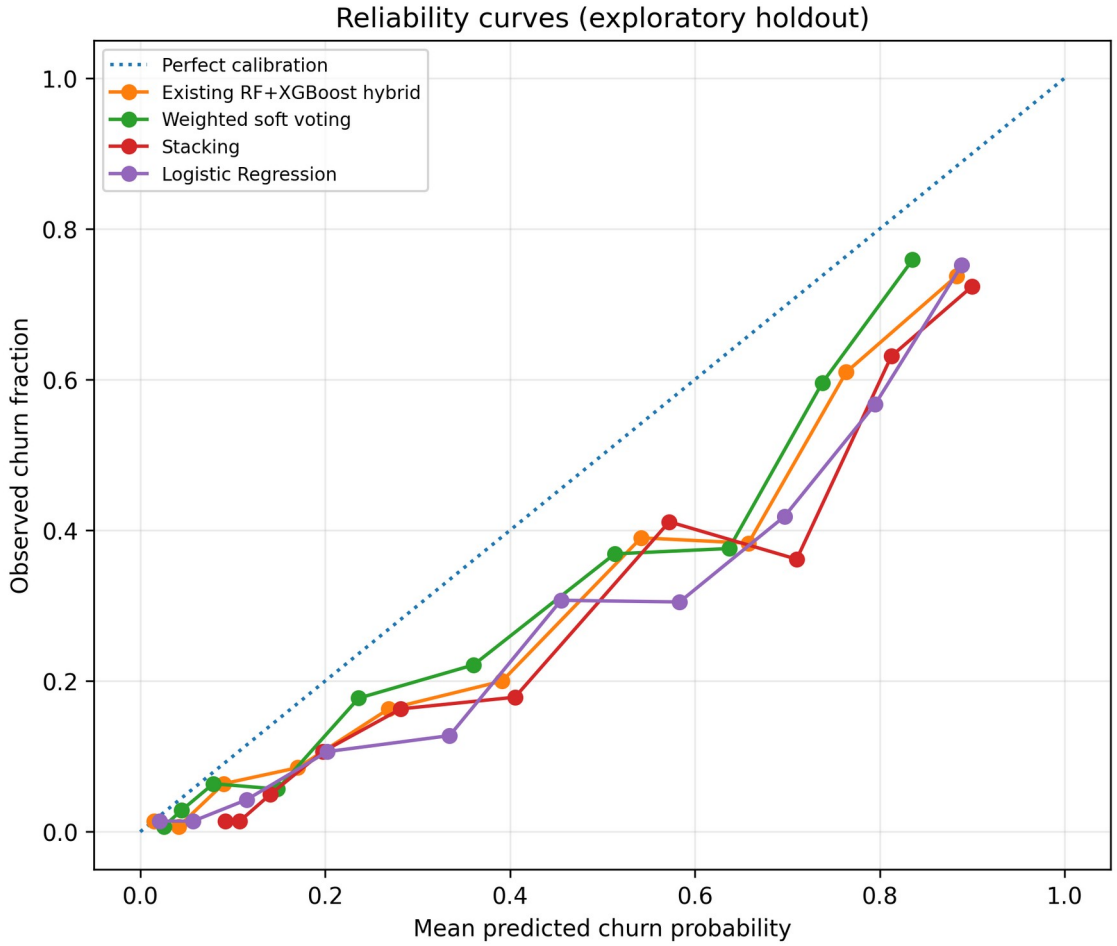


Figure 15. Original project-generated calibration reliability plot.

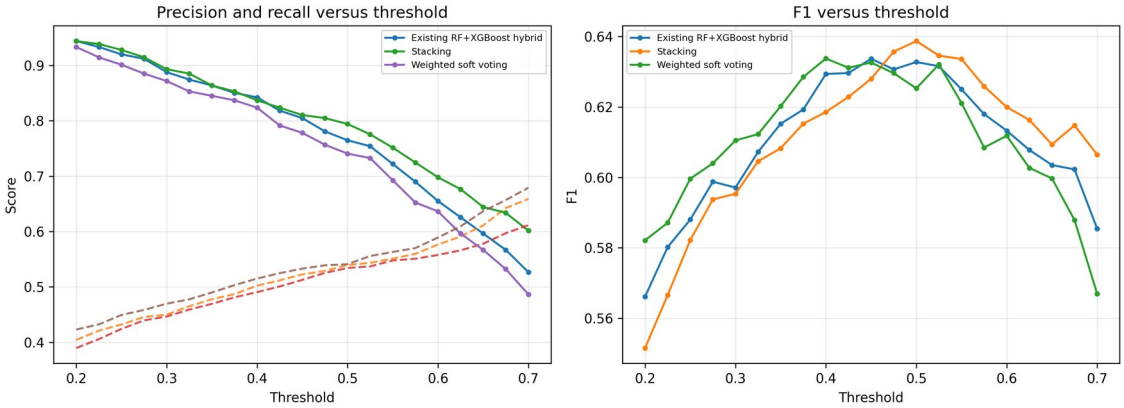


Figure 16. Original project-generated threshold analysis plot.

7.6 Limitations

- The evaluation is based on one public benchmark dataset; performance may differ for other providers, customer populations or time periods.
- The dataset is a static snapshot and does not provide a full time-ordered customer history.
- Class weighting and the fixed 0.5 threshold affect precision, recall and probability interpretation; the threshold is not optimized using retention costs.
- The dashboard screenshots and saved metric files represent recorded execution snapshots. If the project is rerun with different library versions or configuration, results may change.
- SHAP and permutation importance explain model behaviour and feature association, not causality.
- External validation and production monitoring are outside the demonstrated scope.

8. CONCLUSION AND FUTURE SCOPE

The project implements an end-to-end telecom customer churn analytics workflow using the Telco Customer Churn dataset. It includes data cleaning, exploratory analysis, feature engineering, pipeline-based preprocessing, baseline and hybrid model comparison, evaluation, explainability and a Streamlit dashboard with single-record and batch CSV prediction. The dashboard provides several complementary metrics so that the model comparison is not reduced to accuracy alone.

In the supplied dashboard evaluation, the RF–XGBoost hybrid achieves ROC-AUC 0.8434 and F1 0.6344. It is competitive across several measures but does not dominate all metrics. The MLP has higher accuracy and precision, while XGBoost has higher PR-AUC. The result supports the use of a multi-metric evaluation process and careful interpretation of the model’s output. The EDA suggests that contract type, tenure, internet service and payment method are useful segmentation variables for further investigation, but the observed relationships are not causal.

8.1 Future scope

- Calibrate predicted probabilities and assess reliability on a separate validation set.
- Choose decision thresholds using explicit retention costs, contact capacity and expected benefit.
- Perform nested cross-validation and systematic hyperparameter tuning.
- Evaluate on time-based customer records and external telecom datasets.
- Extend fairness and subgroup-error analysis across relevant customer groups.
- Add drift monitoring and a production API with authentication, logging and latency testing.
- Evaluate additional hybrid or stacking candidates under a pre-defined validation protocol.

9. REFERENCES

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Note: The literature register supplied with the project marks some publication metadata for verification. Confirm the full text, author list, issue and article details from publisher records before formal academic submission.

APPENDIX A: ORIGINAL PROJECT FIGURE REGISTER

All visuals in this report are either (a) screenshots supplied by the project team from the running Streamlit application, (b) the original architecture diagram supplied by the project team, or (c) chart files generated and saved by the supplied project code in outputs/figures/. No stock images, AI-generated illustrations or recreated synthetic charts have been inserted.

Figure	Original source used
1, 4	Screenshot 2026-10-25 134305(3).png
2	outputs/figures/01_churn_distribution.png
3	project_architecture_diagram.png (team-supplied diagram)
5	Screenshot 2026-10-25 134418(2).png
6	Screenshot 2026-10-25 134509(2).png
7	Screenshot 2026-10-25 134808(1).png
8	outputs/figures/02_contract_vs_churn.png
9	outputs/figures/03_tenure_vs_churn.png
10	Screenshot 2026-10-25 134425(2).png
11	Screenshot 2026-10-25 134435(2).png
12	Screenshot 2026-10-25 134453(2).png
13	Screenshot 2026-10-25 134525(2).png
14	Screenshot 2026-10-25 134534(2).png
15–16	outputs/figures/calibration_reliability.png; threshold_analysis.png